Asset Performance:

Hea 09 Carbon dioxide sensors

4 Credits

No Minimum Standard

Aim

To encourage the monitoring of internal conditions to ensure a healthy indoor environment is provided.

Question

Are sensors installed in the asset that monitor the levels of carbon dioxide in indoor air?

Credits	Answer	Select a single answer option
0	A.	Question not answered
0	B.	No
2	C.	Yes, in areas subject to large and unpredictable or variable occupancy patterns
4	D.	Yes, in areas subject to large and unpredictable or variable occupancy patterns and in all regularly occupied space

Assessment criteria

Criterion	Assessment criteria	Applicable Answer
1.	Carbon dioxide sensors are installed as follows: a) In mechanically ventilated spaces, the sensors EITHER: • Are linked to the mechanical ventilation system and provide demand-controlled ventilation to the space. OR • Visibly or audibly alert the asset owner or manager or users of the space when carbon dioxide levels exceed the recommended set point. b) In naturally ventilated spaces, the sensors can EITHER: • Visibly or audibly alert the asset owner or manager or users of the space when carbon dioxide levels exceed the recommended set point. OR • Are linked to controls with the ability to adjust the quantity of fresh air, e.g. automatically open windows or roof vents.	C, D
2.	Sensors must be installed, tested, calibrated and maintained in accordance with the manufacturer's instructions. Sensors should be placed to provide	C, D

Criterion	Assessment criteria	Applicable Answer
	representative readings of conditions within each space. Sensors should be wall-mounted and at a height above floor level that corresponds to an average seated or standing head height for the main activity performed within the space.	

Evidence

Criteria	Evidence requirement
-	The evidence below is not exhaustive, please also refer to the 'BREEAM evidential requirements' section in the scope of the Guidance for appropriate evidence types which can be used to demonstrate compliance.
All	Photographic evidence of the sensors.
All	Description of how the sensors work in terms of actuating ventilation or alerting users.
All	Operation and maintenance manuals or records for the installed system(s).

Definitions

Areas with a large and unpredictable occupancy:

The following are examples of these types of spaces:

a) Auditoria

b) Gyms

c) Retail stores or malls

d) Cinemas

e) Waiting rooms.

Occupied space:

A room or space within the asset that is likely to be occupied for 30 minutes or more by an asset user. For the purpose of this issue, the definition excludes the following:

a) Atria or concourses

b) Entrance halls or reception areas

c) Ancillary space, e.g. circulation areas, storerooms and plant rooms.

Additional information

Set points for sensors

Table 16 provides guidance on appropriate set points for carbon dioxide sensors based on the recommendations in the European Technical Report 'PD CEN/TR 16798-2:2019 Energy performance of buildings - Ventilation for buildings - Part 2: Interpretation of the requirements in EN 16798-1 – Indoor environmental input parameters for design and assessment of energy performance of buildings addressing indoor air quality, thermal environment, lighting and acoustics (Module M1-6)'. This information is provided as guidance to inform Assessors and project teams of current industry practice on set points for sensors. The assessment criteria do not specify a target set point in order to allow building or facilities managers to select an appropriate set point for the asset's ventilation system. However, set points should not exceed 1750 ppm and it is recommended that projects aim to achieve the Category II requirements as a minimum.

Table 16: Default design carbon dioxide concentrations

Indoor Environmental Quality (IEQ) Category	Level of expectation	Explanation	Maximum indoor carbon dioxide concentration (ppm)
I	High	For occupants with special needs, e.g. children, elderly, persons with disabilities	950
II	Medium	The normal level used for design and operation	1200
III	Moderate	Will still provide an acceptable environment, but some risk of reduced performance of the occupants	1750
IV	Low	Should only be used for a short time of the year or in spaces with very short time of occupancy	1750

Note: The above values correspond to the equilibrium concentration when the air flow rate is 10, 7 and 4 litres per second per person for Category I, II, and III and IV, respectively; the carbon dioxide emission is 20 litres per hour per person; and assumes an average outdoor air carbon dioxide concentration of 400 ppm.